Homework 4

Probability and Inference foundations

ℹ Instructions

Modify this quarto notebook, change `eval: false` to `eval: true` above, convert it to PDF using typst or to HTML and submit. Total is 70 points

Either HTML or PDF submissions are fine. Please also submit this Quarto file.

{{< downloadthis week-04-student.qmd dname="homework4" label="Download Quarto file" >}}

Rubric:

- Each question is worth 10 points
- Since these are numerical questions with a right answer, full credit will be given for following directions, using appropriate methods and getting the right answer
- Partial credit may be given for incorrect answers if the reasoning and methodology are sound. The correct answer itself is worth 2 points.

ℹ Useful facts

1. If two events A and B are independent, then P(A and B) = P(A) x P(B).
2. The law of total probability states that for two events A and B, P(B) = P(A | B)P(B) + P(A | not B) P(not B)
3. For two events A and B, the conditional probability of B given A is P(B|A) = P(A and B) / P(A)
4. A 95% confidence interval for a mean is given by $(\bar{x} - 2s/\sqrt{n}, \bar{x} + 2s/\sqrt{n})$ where $\bar{x}$ is the sample mean and s is the sample standard deviation from a sample of size n

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Question 1: An air quality monitoring station reports that the probability of a "high pollution alert" on any given day in July is 0.3. Assuming days are independent, what is the probability that exactly 2 out of the next 5 days have high pollution alerts?

Show your work, and confirm your answer using `dbinom(x=2, size=5, prob=0.3)`

💡 Hint

This is an example of the binomial distribution. You can enumerate all the possible sequences with 2 high alerts. The total number of possible outcomes (the sample space) is 2^5.

For the binomial distribution,

$$P(x \text{ events out of } n) = \binom{n}{x} p^x (1-p)^{n-x}$$

where each event is independent and has a chance p of occurring. You can compute ${nx} in using `choose(n, x)` (replacing n and x with appropriate values)

Answer

Question 2: An environmental sensor detects excess ozone (event A) and excess particulate matter (event B). Historical data show: - P(A) = 0.25 - P(B) = 0.40 - P(A∩B) = 0.12 Compute P(B|A) and interpret the result in context. What does it suggest about the joint occurrence of these pollutants?

Are A and B independent? Provide both a numerical check and an interpretation in terms of environmental processes

Answer

Question 3. In a three-region election analysis, probabilities of supporting Candidate Y differ by region:

- Urban (45% of population): 0.70
- Suburban (35%): 0.55
- Rural (20%): 0.40

Find the overall probability that a randomly selected voter supports Candidate Y.

💡 Hint

Use the law of total probability:

$$P(Y) = P(Y \mid \text{Urban})P(\text{Urban}) + P(Y \mid \text{Suburban})P(\text{Suburban}) + P(Y \mid \text{Rural})P(\text{Rural})$$

Answer

Question 4. Suppose a new water-testing kit detects microplastics with 95% sensitivity (true positive) and 90% specificity (true negative). Environmental assessments show that 10% of water samples in the region contain microplastics. If the test gives a positive result, what is the posterior probability the sample truly contains microplastics?

Hint

Use Bayes' theorem

Hint 2

- $P(\text{Positive} \mid \text{Microplastics}) = 0.95$
- $P(\text{Microplastics}) = 0.10$
- $P(\text{Positive}) = P(\text{Positive} \mid \text{Microplastics})P(\text{Microplastics}) + P(\text{Positive} \mid \text{No Microplastics})P(\text{No Microplastics})$

Answer

Question 5: A survey of 1000 voters shows a mean approval rating of 35% for the Governor, with a standard deviation of 12%. Construct a 95% confidence interval for the true approval rating. Interpret your results.

Answer

Question 6. An environmental agency claims that the average nitrate level in groundwater exceeds the safety standard of 10 mg/L. A random sample of 40 wells gives a mean of 10.6 mg/L with a standard deviation of 1.5 mg/L.

a. Formulate the null and alternate hypothesis to test this claim.
b. Construct a simulation experiment to assess the evidence against the null hypothesis and report the p-value from the simulation
c. Interpret this p-value and its implications for the agency's claim. The agency has set a significance level of 3%.
d. Construct a 95% confidence interval for the mean nitrate level.

Note that two people will likely not get the same answers to these questions, due to the random nature of the simulations. However, there is a ballpark of right answers, and you will be graded on your simulation code.

Hint

Follow the examples done in class to set up the simulation experiments. Ensure you're simulating using the information from your null hypothesis.

Answer

Question 7:

The file `dc_sample.rds` is provided you on Canvas. Download this and load this into R with the code

```
dc_data <- readRDS("dc_sample.rds") # specify the right path to the file for
your setup
```

This dataset is a sample of 500 individuals and their race, based on standard Census criteria.

a. Construct a confidence interval of the proportion of black people in DC using this dataset. (5 points)

> **Tip**
>
> You compute a 95% confidence interval for a proportion using the formula
>
> $$\left(\hat{p} - 2 \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + 2 \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right)$$
>
> where $\hat{p}$ is the sample proportion and n is the sample size.

b. Perform a simulation experiment as follows: (10 points)

- Create 1000 datasets of size 500 from this dataset by resampling with replacement.
- For each dataset, compute the proportion of black individuals.
- Construct a distribution of these proportions and report the 2.5% and 97.5% percentiles.
- Compare the confidence interval from part (a) with the distribution from the simulation in part (b).

c. Provide a table showing how often subjects are repeated in one of the resampled datasets. What I'm looking for is a table like the following, where the header denotes the number of times a subject is repeated.

1	**2**	**3**	**4**	**5**
191	86	29	10	2

Also, report on the number of unique individuals included in this resampled dataset (it is **not** 500)

Tip

You can create a single resampled dataset from the original dataset with replacement using the following code:

```
dc_resampled <- dc_data[sample(nrow(dc_data), replace = TRUE), ]
```

or

```
dc_resampled <- dc_data |> slice_sample(p = 1, replace = TRUE)
```

Hint

The easiest way to capture the proportion of black people from the 1000 resampled datasets might be as follows:

```
library(tidyverse)
nsim <- 1000
set.seed(84935)

prop_black <- rep(0, nsim) # It's more efficient to create the vector for
storing results first

for(i in 1:nsim) {
  dc_resampled <- dc_data |> slice_sample(p = 1, replace = TRUE)
  prop_black[i] <- mean(dc_resampled$race == "Black")
}
```

We'll see more efficient and sophisticated methods from the **rsample** package soon.